

An Evaluation of Explanation Methods for Black-Box Detectors of Machine-Generated Text

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Abstract

The increasing difficulty to distinguish language-model-generated from human-written text has led to the development of detectors of machine-generated text (MGT). However, in many contexts, a black-box prediction is not sufficient, it is equally important to know *on what grounds* a detector made that prediction. Explanation methods that estimate feature importance promise to provide indications of which parts of an input are used by classifiers for prediction. However, the quality of different explanation methods has not previously been assessed for detectors of MGT. This study conducts the first systematic evaluation of explanation quality for this task. The dimensions of *faithfulness* and *stability* are assessed with five automated experiments, and *usefulness* is evaluated in a user study. We use a dataset of ChatGPT-generated and human-written documents, and pair predictions of three existing language-model-based detectors with the corresponding SHAP, LIME, and Anchor explanations. We find that SHAP performs best in terms of faithfulness, stability, and in helping users to predict the detector's behavior. In contrast, LIME, *perceived* as most useful by users, scores the worst in terms of user performance at predicting the detectors' behavior.

1 Introduction

Large language models, such as ChatGPT, produce output that is often virtually indistinguishable from human-written text. Their ability to generate human-like text at an unprecedented scale allows for new forms of phishing, disinformation campaigns, and academic fraud (Crothers et al., 2023). Recent work has proposed language-model-based detection methods for machine-generated text (MGT, Solaiman et al., 2019; Guo et al., 2023; Mitchell et al., 2023). These operate as black-box detectors: they provide no explanations for their

decisions. This is insufficient for applications that demand additional evidence, or when wrong decisions affect people, as would be the case for detecting MGT in academia.

To address this, **explanation methods** have been applied to such detectors (Mosca et al., 2023; Liu et al., 2023; Yu et al., 2023), with the majority of papers using SHAP (Lundberg and Lee, 2017) or LIME (Ribeiro et al., 2016). We find that Anchor (Ribeiro et al., 2018), a method that produces rule-based explanations, can also be applied in this setting. All three methods produce local explanations that explain single predictions (MGT or not) by locating relevant elements in the input (words in the document that influenced the prediction). Previously, LIME and SHAP have been used together with detectors of MGT without verifying the quality of the resulting explanations in this setting. However, if such explanations should aid in the categorization of a text as MGT or not, it must be verified that the explanation methods behave as intended. Moreover, reliable explanation methods can generate insights into differences that exist between human- and generated language, while unreliable explanation methods may generate a false sense of understanding of those differences.

In this work, we study explanation methods for detectors of MGT in **three** aspects of quality: Explanations should accurately depict the detector's behavior (**faithfulness**: Jacovi and Goldberg, 2020; Ribeiro et al., 2016; Alvarez Melis and Jaakkola, 2018). They should be sensitive enough and sufficiently deterministic (**stability**: Alvarez Melis and Jaakkola, 2018; Lakkaraju et al., 2020; Nauta et al., 2023), and be effective at communicating the model's decision process to users (**usefulness**: Hoffman et al., 2019; Doshi-Velez and Kim, 2017). To assess how suitable these methods ultimately are for explaining detector decisions, we perform a systematic evaluation with automated metrics and a user study. We test for the above-mentioned aspects

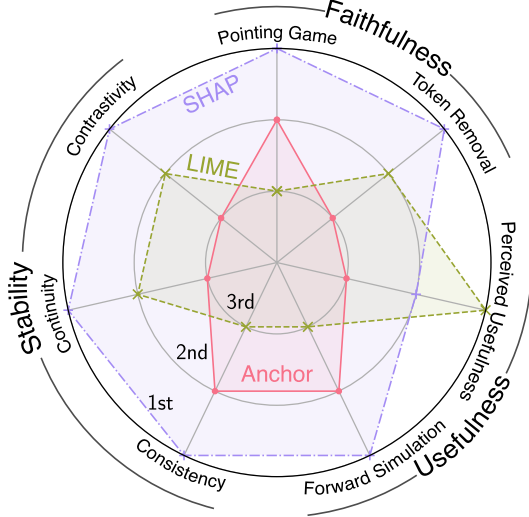


Figure 1: Ranking in the experiments.

with two existing fine-tuned Transformer-based detectors (Guo et al., 2023; Solaiman et al., 2019) and a zero-shot method (Mitchell et al., 2023).

To enable a comparison of faithfulness, we adapt the ideas of a *token removal experiment* (Arras et al., 2016) and the *pointing game* (Poerner et al., 2018) to this task and all explanation methods, and we construct a dedicated test suite to measure the pointing game accuracy in this setting. Stability is assessed with *controlled synthetic data checks* (Nauta et al., 2023) that we construct specifically for the task of detecting MGT. Previous user studies have evaluated usefulness on tasks for which humans already have an intuitive understanding, such as sentiment analysis or income prediction (Ribeiro et al., 2016, 2018; Hase and Bansal, 2020). Identifying patterns in a detector’s behavior is arguably more challenging for users when they have limited knowledge about the features that might be relevant. To keep *forward simulation* experiments feasible for humans in a setting where detectors may utilize complex and unintuitive feature sets, we use a study design similar to that of Hase and Bansal (2020) (originally from the context of movie review sentiment prediction and income prediction) but with a special document selection strategy. Rather than randomly selecting documents, we present users with pairs of documents that we choose based on explanation similarity.

Figure 1 shows an overview of the results. In summary, the contributions of our work are:¹

- (1) The first study to systematically evaluate explanation methods for detectors of MGT, an important basis for understanding differences between generated and natural language; detectors should only be used in critical scenarios if accompanied by explanation methods for which strengths and limitations are experimentally validated.
- (2) Evaluation of all combinations of 3 detectors, 3 explanation methods with 5 automated experiments.
- (3) LM-assisted construction of task-specific test sets to measure contrastivity and faithfulness in this setting.
- (4) In addition to automated metrics, a user study to measure perceived usefulness and actual usefulness in a forward simulation.
- (5) We find that SHAP performs best across all automated metrics. LIME explanations are perceived as most useful by the users but decrease users’ performance at predicting the decisions of the detectors. Neither Anchor nor LIME consistently outranks the other in our experiments.

2 Related Work

Evaluating Faithfulness Arras et al. (2016) evaluate faithfulness with token removal experiments. Note that this setup raises the same concern perturbation-based explanation methods do when applied to text classification: It assumes classifiers behave predictably for partial input.

Controlled synthetic data checks are an alternative type of experiment that foregoes this issue (Nauta et al., 2023). Poerner et al. (2018) also proposed a similar design for evaluating local feature importance explanations for sentiment prediction. We extend both experiments to the task of explaining MGT detection, including rule-based Anchor explanations, in Section 3.1.

Evaluating Stability Quantifying the stability of explanations across runs, *consistency*, with agreement measures is discussed in (Nauta et al., 2023). The properties of *continuity* (similar explanations for similar documents with the same prediction, Alvarez Melis and Jaakkola, 2018) and *contrastivity* (sufficiently different explanations for similar documents but different predictions) described by Nauta et al. (2023) are alternative notions of stability that

¹Code, data and explanations are made available at https://github.com/loris3/evaluation_explanation_quality

rely on convergence to a lesser extent. We propose a setup that generates coherent perturbations from a language model for these tests in Section 3.2.

Evaluating Usefulness Hase and Bansal (2020) perform a *forward simulation* experiment (Doshi-Velez and Kim, 2017), where they assess whether users can predict the detector’s behavior better after being shown explanations. Hase and Bansal conduct their user study with two comparatively simple binary classifiers and tasks (sentiment analysis and income prediction). Given that the explanation methods analyzed here produce outcome- and not model explanations (Guidotti et al., 2018), it is less likely that users will obtain a comprehensive understanding of the model’s behavior or relevant features and successfully apply that to new instances of human-written and MGT. We propose a strategy to increase the feasibility of this kind of experiment in Section 3.3. Besides measuring user performance, we also assess *perceived usefulness*. This is measured in a *rating task* with questions adapted from Hoffman et al. (2019).

3 Defining Explanation Quality Metrics for Detectors of MGT

We systematically evaluate explanation quality along the axes of faithfulness, stability, and usefulness. Throughout this section, $f(d_i)$ refers to the decision of a detector f for a document d_i . D is the base dataset of human-written and machine-generated documents $\{d_i\}$. SHAP and LIME provide *feature importance scores* for tokens from the input document d_i as explanations for decisions made by f (Lundberg and Lee, 2017). Those tokens that had the strongest impact on the detector’s decision should be attributed the highest scores. Anchor explanations are expressed as *if-then rules* (Ribeiro et al., 2018). There can be multiple valid Anchors for a document. Ribeiro et al. (2018) define an Anchor $A_{i,m}$ for the document d_i as a set of tokens $A_{i,m} = \{t_{i,j}\}$ that, if present in the document, guarantee the same decision as for d_i with a probability greater than τ for perturbations in the local neighborhood of d_i .

3.1 Evaluation of Faithfulness

A faithful explanation method should accurately depict the detector’s behavior (Jacovi and Goldberg, 2020; Ribeiro et al., 2016; Alvarez Melis and Jaakkola, 2018). The first test for this is a **pointing game** similar to Poerner et al. (2018). Random sen-

tences from the base dataset D are concatenated to form a synthetic dataset of *hybrid documents* $D^h = \{d_i^h\}$. The length of these hybrid documents is set to match the mean document length by the number of sentences in the original dataset in our setup. It is assumed that the detector’s decision $f(d_i^h)$ on such a document is based on segments that were originally part of documents with ground truth y_i equal to $f(d_i^h)$. A faithful explanation method should hence find these segments to be more important for the decision than those originating from opposite-class documents. In the pointing game, feature importance explanation methods are therefore awarded hits for a document d_i^h if the token with the highest feature importance score $t_{i,max} \in d_i^h$ was in fact originally part of a document with ground truth y_i equal to $f(d_i^h)$ (Poerner et al., 2018): $S_x = \bigcup_{\forall d_i \in D} \{t \in d_i \mid y_i = x\}$, $hit(d_i^h, t) = \mathbb{1}[t \in S_{f(d_i^h)}]$. Where $\mathbb{1}[\cdot]$ is 1 if $\cdot$ evaluates to true, 0 otherwise.

The *pointing game accuracy* is the fraction of documents in the dataset D^h that get awarded hits. For feature importance type explanations, it is given as: $Acc_{pg} = \sum_{\forall d_i^h \in D^h} hit(d_i^h, t_{i,max}) / |D^h|$.

Poerner et al. (2018) only evaluate feature-importance explanation methods. Note that there is no distinction by importance between tokens within an Anchor. Furthermore, a single Anchor can span multiple sentences. We therefore attribute hits proportionally for this explanation method. The hit function then tests all tokens $t_{i,j}$ specified by the Anchor A_i individually and returns the average number of hits instead of a binary value:

$$hit^R(d_i^h) = \frac{\sum_{\forall t_{i,j} \in A_i} hit(d_i^h, t_{i,j})}{|A_i|}.$$

A **token removal experiment** as in Arras et al. (2016) is performed as the second test for faithfulness. Let d_i be the original document and d_i^k a version with the k top-tokens by feature importance towards $f(d_i)$ removed. Arras et al. plot the accuracy of the detector at different k with respect to the ground truth y_i of the original documents. For Anchor explanations, we remove tokens from the Anchor that applies to the highest proportion of documents in the local neighborhood (the one with the highest τ) in random order.

3.2 Evaluation of Stability

We consider three different notions of stability. The individual experiments are based on the principles of consistency, continuity, and contrastivity as char-

acterized in Nauta et al. (2023).

Consistency We measure consistency across five explanations for the same document with the agreement metric Krippendorff’s α (Krippendorff, 1970), calculated on the explanations’ feature-importance vectors.²

Continuity The continuity between the explanation for the original document and explanations for a set of 5 perturbations $\{d_i^k\}$ is also measured with Krippendorff’s α . For each document d_i from the original dataset, a single token is randomly selected and replaced with an arbitrary number of tokens using the T5 language model (t5-small 60.5M params; Raffel et al., 2020). We set the maximum output length to 150 tokens. It is verified that $f(d_i^k) = f(d_i)$. In some instances, T5 fails to generate five unique replacements. In these cases, the token is replaced with a random token from the vocabulary.

Contrastivity The high-level goal of contrastivity evaluation is to verify that documents that are similar in content, but are assigned different labels by the detector, get sufficiently different explanations (Nauta et al., 2023). We implement this principle by editing documents from the dataset with a language model in order to obtain pairs of coherent documents (d_i, d_i^{Ω}) where d_i and d_i^{Ω} are similar but get assigned different labels. In each iteration k , one token is deleted from the end of the document. The shortened document d_i^* is used as an initial prefix for the autoregressive foundation model facebook/opt-350m (Zhang et al., 2022) to generate a continuation that (together with the prefix) will constitute the perturbation d_i^{Ω} . We halt removing tokens as soon as $f(d_i^{\Omega}) \neq f(d_i)$.

To increase the chance of a label flip with smaller edits, five attempts are made at every k . Still, if the first tokens in the document are highly influential for the decision of the detector, more tokens need to be deleted until a label change can be observed, resulting in more dissimilar perturbed documents.³ To retain a certain level of similarity between documents, perturbations that edit more than 50% of the original document are discarded. The resulting synthetic datasets of pairs (one set per detector) are further described in Figure 11 in Appendix D.

Two scores are subsequently calculated on the explanations for d_i and d_i^{Ω} . These verify whether the explanations are consistent with the genera-

tion strategy: Given that the left part of d_i and d_i^{Ω} are identical, but $f(d_i^{\Omega}) \neq f(d_i)$, one expects the filled-in part in d_i^{Ω} to have had a strong influence on the detector when labeling d_i^{Ω} . A sufficiently sensitive explanation method should be able to detect this.

The scores are formulated as synthetic data checks like the pointing game described above. c_{inter} is calculated on the parts that differ across the two documents, d_i^- and $d_i^{\Omega-}$. It tests whether the mean feature importance score towards $f(d_i^{\Omega})$ (denoted $\mu_{\vec{v}}^{\Omega}(\cdot)$) is higher for $d_i^{\Omega-}$ than it is for d_i^- . The hit function for this case is: $hit_{inter} = \mathbb{1}[\mu_{\vec{v}}^{\Omega}(d_i^{\Omega-}) > \mu_{\vec{v}}^{\Omega}(d_i^-)]$.

c_{intra} shares the same intuition but is defined only on d_i^{Ω} . As the left parts of d_i and d_i^{Ω} are identical, but $f(d_i^{\Omega}) \neq f(d_i)$, one expects the generated part $d_i^{\Omega-}$ to have a higher average feature importance score towards $f(d_i^{\Omega})$ than the shared part: $hit_{intra} = \mathbb{1}[\mu_{\vec{v}}^{\Omega}(d_i^{\Omega-}) > \mu_{\vec{v}}^{\Omega}(d_i^{\Omega*})]$.

Both scores are again given as the fraction of documents in the dataset that score a hit: $c_{\{intra,inter\}} = \frac{\sum_{\forall(d_i, d_i^{\Omega})} hit_{\{intra,inter\}}(d_i, d_i^{\Omega})}{|D|}$.

3.3 Evaluation of Usefulness

We define usefulness as the explanation method’s ability to improve users’ understanding of the detector’s behavior. To quantify this, we modify the design for a forward simulation experiment in Hase and Bansal (2020). In phase 1 of the experiment, users inspect decisions of the detector on a set of documents $\{a_j\}$ without explanations. In phase 2 they are then instructed to anticipate the detector’s decision (not to guess the true document class) on a second set $\{b_j\}$. Phase 3 provides explanations for set $\{a_j\}$ (Figure 2). Users conclude the experiment by labeling the documents from set $\{b_j\}$ again in phase 4. The change in user accuracy from phase 2 to 4 is reported as a measure of performance.⁴ We ask three additional questions from Hoffman et al. (2019) for every explanation shown in phase 3 to measure *perceived usefulness*. These are provided in Appendix C. In order to be able to track whether users can pick up regularities between the different phases, we apply a document selection strategy that ensures a minimal overlap of features in the documents, see Appendix B.

²Anchor explanations are one-hot encoded: If a token is part of an Anchor, it is deemed relevant.

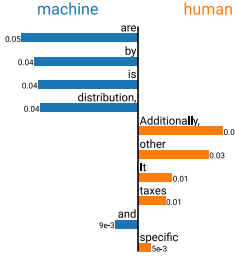
³E.g., documents that start with greetings like "Hi!".

⁴See Figure 4 in Appendix B for an overview of all phases.

The detector **correctly** predicted that this document is **machine generated**.

Detector output:

p(machine) = 100%
p(human) = 0%



The price of fuel in Australia, like in any **other** country, is determined **by** a variety of factors, including the cost of production, transportation, and **distribution**, as well as market demand and supply. **Additionally**, **taxes** and other government policies can also impact the price of fuel. **It is** difficult to say whether the current price of fuel in Australia is too high or not without more information about the specific market conditions and factors that **are** influencing it.

I'm not familiar with the term "cake and underwear" stocks. It's possible that this is a slang term used by some investors to refer to stocks that are considered to be essential or basic necessities, similar to how cake and underwear are essential items that people need on a regular basis. However, without more context it is difficult to say for certain what this term refers to. It's possible that it could be used to describe a variety of different types of stocks, depending on how it is being used.

I think the detector predicted...



Figure 2: Information shown for LIME in phase 3 (left); and in the annotation phases 2 and 4 (right).

4 Technical and Experimental Details

4.1 Dataset

A subset of the H3 dataset by [Guo et al. \(2023\)](#) is used for the experiments. They add ChatGPT written answers to question answering datasets from various domains.⁵ We remove all questions from WikiQA and ELI5, as the source datasets contain crawling artifacts that make identifying human texts trivial. Moreover, only documents that are between 50 and 150 words long are kept. This is to ensure a sufficient input length for the zero-shot detector. In the end, 1016 documents remain. Due to the high computation time involved with generating explanations, only 30% of documents (N=305) are used (stratified split of machine-generated and human-written documents) for our experiments. As discussed in Section 3, we derive additional synthetic datasets from this set.

4.2 Detectors and Explanation Methods

We apply the explanation methods to three **detectors** of machine-generated text: The fine-tuned RoBERTa model of [Solaiman et al. \(2019\)](#), the model of [Guo et al. \(2023\)](#) (which uses the same base model, but has been trained on the full H3 dataset), and the zero-shot model of [Mitchell et al. \(2023\)](#), see Appendix A. We perform no further fine-tuning.

Feature Importance Explanation Methods assign scores to individual tokens in the document with the goal to quantify the effect a token has on the decision of the classifier. **LIME** trains a local surrogate model on a set of data point perturbations and corresponding detector outputs ([Ribeiro et al.,](#)

2016). The number of perturbations to use for fitting this model, and its size, have to be chosen manually. The number of samples was set to 1k for the RoBERTa-based detectors and 500 for the zero-shot method to match SHAP’s runtime.⁶ The default number of 10 features to show as an explanation appears appropriate for the document length used here (50-150 tokens). The implementation of **SHAP** we use (Partition Explainer: [Lundberg and Lee, 2017](#), default parameters) computes Owen values ([Owen, 1977](#)) as a measure of feature importance.

Rule-based explanation methods – like **Anchor**, explain complex decision processes with short rules ([Ribeiro et al., 2018](#)). Among valid rules as outlined in Section 3, Anchor’s search algorithm attempts to select those that apply to the highest proportion of perturbations in the local neighborhood. In its default implementation, Anchor is often unable to terminate for a single document within an hour of runtime with these detectors. We employ the following strategies to make the computation of Anchors feasible here: We choose a low target level of precision ($\tau = 0.75$) and impose a limit on the number of samples used during construction (200 samples per candidate Anchor). For generating perturbations, we employ Distill-RoBERTa (distilroberta-base 82.8M params; [Sanh et al., 2020](#)) instead of DistillBERT, to increase their coherence and edit at most 20% of tokens per perturbation. Combined, this reduces the computation time to an average of roughly 5 minutes per explanation for the RoBERTa-based detectors and 15 minutes for DetectGPT on the original documents.

Perturbation Strategy An experiment that tests

⁵ELI5 ([Fan et al., 2019](#)), WikiQA ([Yang et al., 2015](#)), FiQA ([Maia et al., 2018](#)), Medical Dialog ([He et al., 2020](#)) and Wikipedia computer science articles ([Guo et al., 2023](#)). Published under the CC-BY-SA licence.

⁶Resulting in an average of ≈ 30 s for the RoBERTa-based detectors and ≈ 415 s for DetectGPT per explanation.

the effect of different perturbation strategies (Appendix D) did not single out a method that works equally well for all detectors. In the interest of consistency, perturbations are generated by replacing tokens with the mask token of the detector’s tokenizer for both LIME and SHAP throughout all experiments. Anchor offers masking with a specified token or to perturb with a language model. The latter strategy was used, as it was found to terminate considerably faster with these detectors.

4.3 User Study

We recruited 36 participants (B.Sc., M.Sc., and PhD students) with a background in computer science and English reading proficiency at the C1 CEFR level or higher. Of them, 27 stated that they had never worked with explanation methods before. The instructions shown to users on the individual explanation methods are provided in Appendix C. We offered a compensation of €10 for their participation. The experiment was conducted online through a purpose-built web service.

5 Results

Aggregate results per explanation method for faithfulness and stability are reported in Table 1, those from the user study in Table 2. See Appendix D for results per detector-explanation method pairing.

5.1 Faithfulness

In the **pointing game** (Table 1), SHAP performs best and a series of binomial tests (H_0 : No difference between one method and the next best in the ranking) shows that the difference between SHAP and Anchor, as well as the one between Anchor and the random baseline are significant ($p < 0.05$). LIME does not outperform a baseline of random feature importance scores but is also not significantly worse ($p=0.143$).

For the **token removal experiment**, results for initially correct predictions ($f(d_i) = y_i$) and initially wrong predictions ($f(d_i) \neq y_i$) are plotted separately in Figure 3 to allow for consistent interpretation of accuracy scores (Arras et al., 2016). We report the change in accuracy at $k = 10$ tokens removed ($\Delta_{k=10}$) in Table 1, corresponding to the maximum number of tokens Anchor and LIME include in their explanations. We provide the mean scores for five explanations with random feature importance vectors as a baseline. For $f(d_i) = y_i$, the most important features are removed first. A faithful (feature importance) explanation method should

have a steep drop in accuracy. For $k < 10$ tokens removed, the accuracy of LIME drops slightly faster than that of SHAP. The average accuracy drops below 50% for SHAP at roughly 10 tokens masked. For Anchor, the accuracy drops slower than for all other methods.

For $f(d_i) \neq y_i$, one expects the accuracy to increase. Note that few initially wrong examples are available (105 vs. 810 initially correct cases) given the high accuracy of two detectors. The average accuracy of LIME and SHAP does not increase faster than the random baseline here. Anchor flips the label in all instances and thus archives a perfect score at $k = 10$ for initially wrong examples.

5.2 Stability

Consistency For Krippendorff’s α , a score of 0 reflects an agreement by chance, 1 perfect agreement, and negative values systematic disagreement (Krippendorff, 1970). Scoring 0.136 and 0.160 respectively, LIME’s and Anchor’s consistency in re-runs is far below what could be considered reliable. SHAP (Partition Explainer) is deterministic for the detectors of Guo et al. (2023) and Solaiman et al. (2019). Note that the zero-shot detector of Mitchell et al. (2023) is not deterministic. We include this detector in the aggregate scores in Table 1 for consistency with the other experiments. The ranking remains unaffected by this, per-detector results are provided in Appendix D.

Continuity SHAP’s explanations remain stable under small perturbations ($\alpha > 0.8$ for the detectors of Guo et al., 2023 and Solaiman et al., 2019). Those from LIME and Anchor do not, but agree with each other better than explanations from the random baseline.

Contrastivity SHAP ranks highest in both synthetic data checks c_{inter} and c_{intra} : It can identify that the filled-in part is responsible for flipping the label more reliably than LIME and Anchor.

5.3 Usefulness

Forward Simulation Table 2 shows the change in user accuracy from phase 2 to phase 4 and the results from McNemar’s tests. SHAP is the best-performing method, followed by Anchor and LIME. The increase in user accuracy for both SHAP and Anchor is not significant at $p < 0.05$. However, participants who have seen LIME explanations perform 13.12% worse after being shown explanations than they did before ($p=0.006$).

	Pointing Game	Token Removal		Consistency	Continuity	Contrastivity	
	Acc_{pg}	$\Delta_{right, k=10}$	$\Delta_{wrong, k=10}$	α	α	C_{inter}	C_{intra}
Random	0.565	51.2%	75.1%	-0.167	-0.139	0.486	0.498
LIME	0.546	46.9%	57.3%	0.136	0.394	0.598	0.587
Anchor	0.589	28.3%	100.0%	0.160	0.210	0.512	0.289
SHAP	0.692	50.4%	62.3%	0.695	0.596	0.799	0.774

Table 1: Results for the faithfulness and stability experiments. The best scores are highlighted in **bold**.

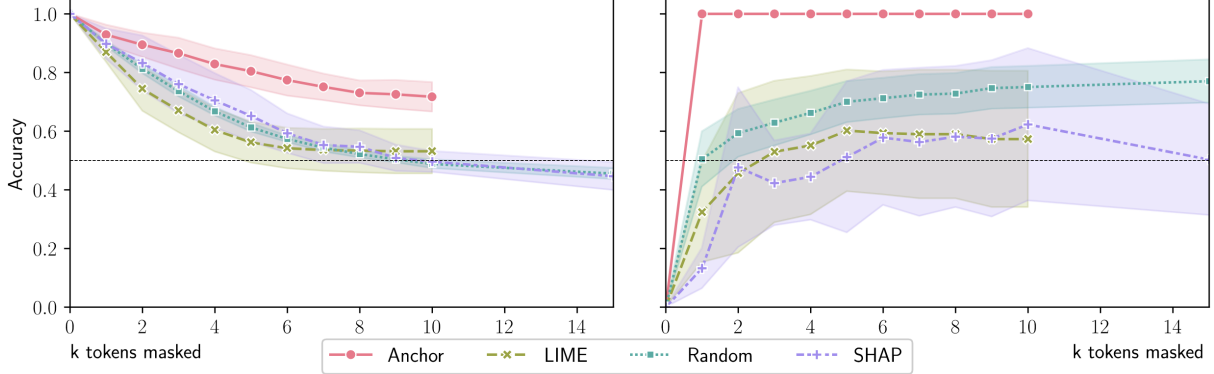


Figure 3: Accuracy at k tokens masked. A faithful explanation method should feature a steep decline (initially correct predictions, left) or steep incline (initially wrong predictions, right). Only SHAP explanations cover more than 10 tokens. Mean across all detectors, error bars at ± 1 standard error.

	Forward Simulation				Perceived Usefulness		
	Without	With	Change	p	Q1: <i>Why</i>	Q2: <i>How</i>	Q3: <i>Helpful</i>
LIME	0.741	0.644	-13.12%	0.006	3.60	3.37	3.31
Anchor	0.694	0.699	0.67%	1.000	2.57	2.48	2.51
SHAP	0.755	0.778	3.07%	0.551	3.06	2.86	2.84

Table 2: Results from the user study. 5-point Likert scale (3 = neutral, 5 = strongly agree).

Perceived Usefulness Conversely, users rated LIME best, SHAP second, and Anchor third across all three questions assessing perceived usefulness. For SHAP, users only tended to agree with the first item, where they were asked whether they could *understand why* the detector decided the way it did from an explanation (Figure 5 in Appendix C). They disagreed with the two other items, asking whether they better *understood how* the detector works and whether they thought the information from the explanation *would help them* perform better in the second round of annotation. They tended to agree for LIME with all three statements, for Anchor to disagree on average.

6 Discussion

Faithfulness The explanations from SHAP and Anchor are in line with the data generation strategy in the pointing game, and LIME achieves the lowest pointing game accuracies. The pointing game and the token removal experiment are based on different assumptions: The pointing game rewards if

the explanation method is consistent with the assumption that a classifier relies on material that is associated with a predicted class more than with unrelated material. This is a reasonable assumption, especially for classifiers with much better than random accuracy. The token removal experiment rewards if the explanation method ranks features on top that if removed in a greedy fashion influence prediction the most. This might be at odds with other notions of importance, such as Shapley values, that estimate importance averaged over different combinations of features.

The definition of an Anchor states that changing tokens not part of it should not affect the detector’s prediction. Therefore, one would expect that masking tokens which are part of an Anchor would change the prediction more frequently than random masking. One should, however, not expect the change in accuracy to be more pronounced than for the other methods: Anchor does not attempt to identify a set of tokens that affect the decision most. It merely aims to provide a set that is *important*

enough to cause a certain outcome. While Anchor is less accurate in the token removal experiment for the majority of cases (where the detectors predict correctly), it however outperforms the other methods for the much smaller class of initially wrong predictions.

Stability The partition tree used by SHAP’s Partition Explainer is obtained with a deterministic algorithm. SHAP is therefore able to accomplish a perfect score in the **consistency** experiment for the two deterministic detectors (in contrast to LIME that uses a different random seed and set of perturbations for each explanation attempt). The difference between SHAP, LIME, and Anchor is less pronounced in the **continuity** experiment, which measures how explanations are affected by small changes to the input documents. Regarding **contrastivity**, where we assess whether explanations for similar documents with different predictions are sufficiently different, SHAP is more performant than LIME and Anchor: SHAP attributes importance in the expected way more often than the other methods (c_{inter}). SHAP is also considerably more successful at identifying that the filled-in part, and not the shared part, is responsible for flipping the label (c_{intra}).

Based on the results from these three experiments, SHAP appears to be the most sensitive method and to produce more stable explanations than LIME and Anchor in our setup.

Usefulness When measuring the usefulness of explanations presented to users, different aspects must be differentiated: a (partial) understanding of the model to be explained (the detector in our case) vs. the perceived degree of understanding of the model. These two may not be the same, it can be that an explanation model gives plausible explanations that do not provide actual insight into the detector’s behaviour. Another subtle distinction is that between (perceived or actual) understanding of model behaviour vs. understanding of task characteristics (what makes human and generated texts different). We have tested increased model understanding in the forward simulation experiment, where we measure the effect of seeing predictions without and with explanations on the ability to anticipate model behavior on new inputs. We also reported *perceived* usefulness of the different explanations. However, even though we carefully phrased the questionnaire to inquire the perceived ability to understand model behaviour (*I now better understand how the detector works*), it is conceiv-

able that users conflated that with a better understanding of task characteristics.

Our results show that none of the methods substantially increased the users’ ability to anticipate detector predictions (with SHAP and Anchor having a small positive but not statistically significant effect). One reason could be that in contrast to simpler tasks for which feature-based explanation methods had been applied to before (e.g., binary sentiment prediction), the possible feature space is much larger for the MGT detection task (obvious sentiment-bearing words vs. subtle frequency choices or constructions). Good performance in the automated metrics (faithfulness and stability) did not translate to usefulness in the forward simulation. The method with the best perceived usefulness (LIME) shows the worst performance there (as well as in many of the automated metrics). Therefore, our study is a strong motivation to rethink the applicability of local feature-based explanations to complex tasks, and highlights the importance of accompanying automated evaluation of explanation methods with user studies.

7 Conclusion

In this work, we conduct the first evaluation of explanation methods for detectors of machine-generated text. We find that SHAP fulfills the theoretically motivated properties of good explanations of faithfulness, stability, and usefulness best in our experiments. However, no explanation method led to a significant increase in performance in a user study that tested users’ ability to predict the behavior of detectors with the help of explanations. For LIME, users’ perceived usefulness of explanations did not align with the measured performance, nor with the results from the faithfulness and stability experiments.

Given the results from the user study, we advise against implementing combinations of these methods and detectors in their current form into systems that face untrained users. We do however see their exploratory value for model- or dataset inspection. Based on the results from the faithfulness and stability experiments, we recommend SHAP for this application.

8 Limitations

We restrict our study to explanation methods that can be applied to black box predictors of machine-generated text, i.e., methods that generate expla-

nations by tracing causal effects from manipulating inputs and observing the corresponding predictions. These methods are generally applicable to all detectors of generated text, and do not have access to training data, activations, attention patterns, and other internal states or detector-specific information. SHAP and LIME, the most prominent explanation methods for classifiers, fall into this category. Such black-box explanations allow for comparisons across detectors, because they do not depend on detector-specific properties. On the flip-side, explanation methods for black-box predictors rely on input-perturbations, and model-specific methods (such as attention patterns) or explanations that trace back training-data influence are not included in our study.

More generally, most current explanation methods only provide explanations by identifying prediction regularities or causal structures of the predictor, but do not build a model of the human users to which the explanations are shown, and therefore may not be considered full explanations according to the theory of mind (i.e., they do not take into account what can be expected to be already known by a user). However, we argue that faithful, stable and useful feature-importance explanations will be an important building block for future explanation setups that combine such explanation algorithms with an elaborate model of the system context and a tailored user interface, and which will include a theory of mind of individual system users.

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A Appendix: Detectors

Guo (Guo et al., 2023) With their dataset, Guo et al. also ship a fine-tuned RoBERTa model (roberta-base 125M params; Liu et al., 2019). The accuracy of this detector on the sub-set for which explanations were generated is 0.99.

Solaiman (Solaiman et al., 2019) This detector was fine-tuned on the Webtext and GPT-2 output datasets (Solaiman et al., 2019), but uses the same base model as Guo et al. (2023). It is therefore treated as an out-of-distribution detector. Its accuracy is 0.92.

DetectGPT (Mitchell et al., 2023) This is a zero-shot method. It is set up here with a smaller language model (pythia-70m; Biderman et al., 2023) as suggested by Miresghallah et al. (2023) and accomplishes an accuracy of 0.74. This, and reducing the number of perturbations per evaluation from 100 to 5 is done to reduce inference time from roughly 6.3 (15.8 with GPT-2) to 0.9 seconds per document.⁷ A comparison with the original implementation is provided in Appendix D.

B Appendix: User Study

Document Selection Users can only apply observations about detector behavior in the annotation phases (2 and 4) if they have seen similar cases in the teaching phases (1 and 3). Rather than using random documents, we construct two sets $\{a_j\}$ and $\{b_j\}$. Each document a_j , to be shown in phases 1 and 3, has a corresponding document b_j shown in phases 2 and 4. The explanations for the documents in a pair (a_j, b_j) should be *sufficiently similar* so that the task is feasible for human annotators.

In the first step, all possible combinations of documents in the dataset are sorted by their cosine similarity in a bag-of-words encoding of the most salient features (as in Ribeiro et al., 2016). A second step aims to maintain sufficient diversity across pairs: Among the top-k most similar pairs, the n pairs that maximize coverage are chosen, defined here as the number of features with a non-zero importance score in the global encoding.

For rule-based explanations, this strategy cannot be applied, as we lack an appropriate similarity metric for this type of rule. Pairs are found by testing for set-equality: For a given document a_j , all other documents in the dataset that share an Anchor with a_j , and have $f(d_i) = f(a_j)$ are collected. Anchors

can be as short as one token long, and documents therefore might only overlap in one word with this strategy. A second step addresses this: If multiple documents share an Anchor with a_j , only the document with the highest Jaccard index with a_j is kept. This is done so that the two documents share as similar of a context as possible. Finally, n pairs are selected at random from these candidates.

Note that these selection strategies will yield different sets for different pairings of detectors and explanation methods. The experiment are conducted with three sets of pairs, one per detector (Figure 4). Of the 18 pairs in a set, 6 will be determined by each explanation method. Document- and explanation similarity for the three sets are reported in Table 3. The datasets are balanced in respect to the detectors’ predictions, to eliminate counting examples in phases 1 or 3 to infer the number of machine- and human-documents in phases 2 or 4 as a strategy (Hase and Bansal, 2020).

Assignment In total, there are 9 different explanation method-detector combinations. One user only sees explanations from one explanation method for one detector. Users who are assigned the same detector, but different explanation methods, see the same set of documents. Those who are assigned the same explanation method, but different detectors do not.

Evaluation We report the average change in user accuracy per explanation method. Whether showing explanations leads to a significant increase in performance is assessed with McNemar’s test (McNemar, 1947).

Table 3 reports the document- and explanation similarity of pairs obtained with the proposed selection strategy. Pairs of explanations are significantly more similar than random selections, while not featuring overly similar documents.

⁷The default setting for this detector would require 10^5 generations with GPT-2 for a single explanation.

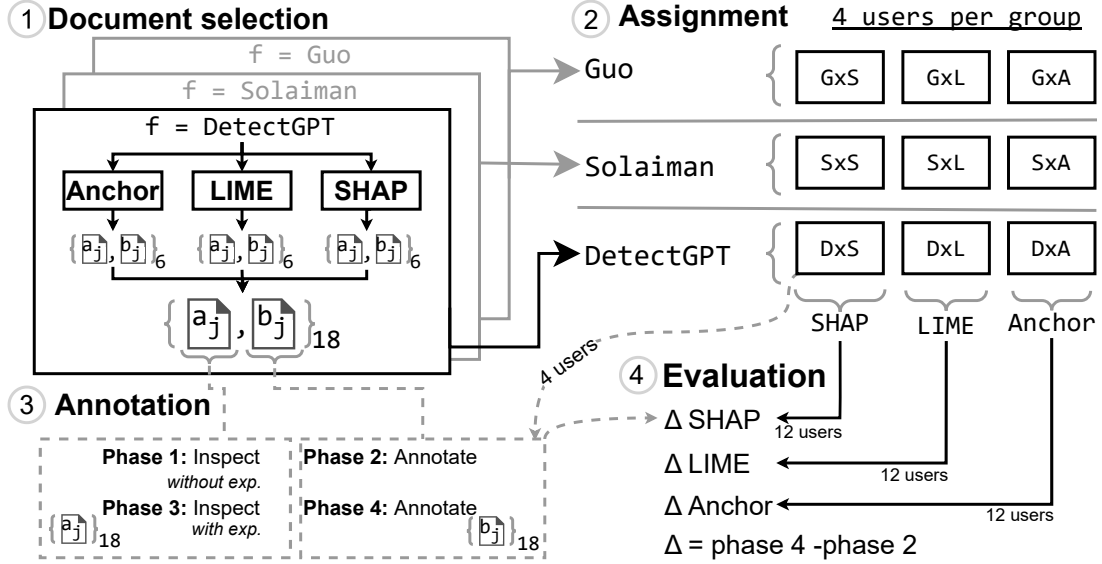


Figure 4: Setup of the user study

(a) Document Similarity					(b) Explanation Similarity				
	Set	Method	Random	Increase		Set	Method	Random	Gain
Jaccard	Solaiman	0.17	0.12	0.05	Cosine Sim FI-Features	Solaiman	0.31	0.17	0.15
	Guo	0.14	0.12	0.02		Guo	0.38	0.22	0.16
	DetectGPT	0.14	0.11	0.03		DetectGPT	0.24	0.11	0.14
Cosine TF-IDF	Solaiman	0.12	0.08	0.05	# Matching Anchors	Solaiman	0.33	0.09	0.24
	Guo	0.12	0.09	0.04		Guo	0.89	0.42	0.47
	DetectGPT	0.12	0.08	0.04		DetectGPT	0.28	0.06	0.22

Table 3: Similarity between pairs (a_j, b_j) in the datasets for the user study against the mean similarity of 10 random selections ($p < 0.05$ bold). Cosine similarity is given as the mean of SHAP and LIME

C Appendix: Annotator Guidelines and Examples

For each document, please also rate to what extent you agree with these statements:

Q1 From the explanation, I **understand why** the detector decided the way it did for this document.

Select *agree* or *strongly agree* if you think the visualization presents sufficient evidence for why the detector decided the way it did in this specific case.

Select *disagree* or *strongly disagree* if you can't figure out why the detector decided the way it did.

Q2 From the explanation, I now better **understand how** the detector works.

Select *agree* or *strongly agree* if you think this explanation increased your understanding of how the detector reasons.

Select *disagree* or *strongly disagree* if you don't.

Q3 The information from this explanation **will help me** predict the detector's behaviour.

Select *agree* or *strongly agree* if you think you could apply this information to the documents you labelled in the previous phase. You will do so in the next phase.

Select *disagree* or *strongly disagree* if you don't.

Figure 5: Instructions for the Likert scale items shown in Phase 3 adapted from [Hoffman et al. \(2019\)](#)

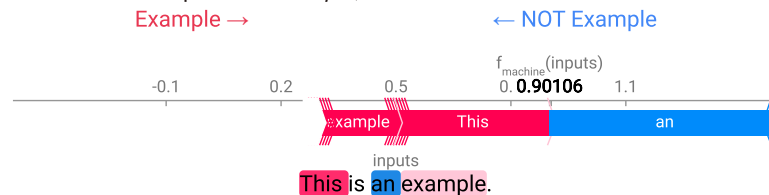
This method attempts to highlight words that are important for the detector's decision.

Suppose you have a detector that detects *examples* (it assigns the labels *example* or *not an example*).

For the document

This is an example.

where the detector predicts *example*, the visualization can look like this:



Each word is assigned an **importance score**.

The words highlighted in pink increase the detector's confidence that this is an *example*. The words highlighted in blue decrease it. The darker the shade, the stronger the effect.

The visualization depicted here can be interpreted as follows:

The word *This* appears to be more important for the detector's decision than the word *example* is.

The word *an* decreases the detector's confidence. All other words increase it.

The upper part of the plot provides the same information: the word in blue lowers the confidence that this is an *example*, the words in pink increase it.

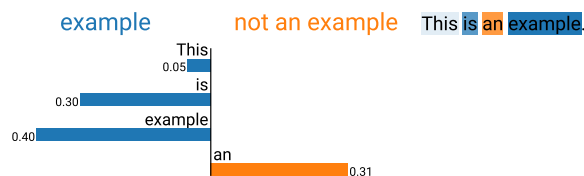
This method attempts to highlight words that are important for the detector's decision.

Suppose you have a detector that detects *examples* (it assigns the labels *example* or *not an example* to documents).

For the document

This is an example.

where the detector predicts *example*, the visualization can look like this:



Each word is assigned an **importance score**.

The words highlighted in blue increase the detector's confidence that this is an *example*. The words highlighted in orange decrease it. The darker the shade, the stronger the effect.

The visualization depicted here can be interpreted as follows:

The word *example* appears to be more important for the detector's decision than the word *This* is.

The word *an* decreases the detector's confidence. All other words increase it.

Figure 6: Instructions on SHAP and LIME shown in Phase 3

This method provides you with **rules** in an attempt of explaining the detector's decision.

Suppose you have a detector that detects *examples* (it assigns the labels *example* or *not an example* to documents).

For the document

This is an example.

where the detector predicts *example*, a rule can look like this:

Example	▽ Examples where the Detector predicts example
This is an example.	< 1 2 3 > This is the example.
Explanation of detector prediction	▽ Examples where the Detector DOES NOT predict example
If ALL of these words are in the text: ✓ example ✓ This ✓ is	< 1 2 > This is not an example.
The Detector will predict example 90.0% of the time	

The box on the upper right specifies **conditions** that likely led to the prediction being *example*.

These were generated by tracking the detector's decisions on **similar documents**, some of which are provided in the boxes at the bottom.

The rule depicted here can be interpreted as follows:

Of all analyzed documents, those that contained all the specified words got the prediction *example* in 90% of cases.

In other words, if one were to change out any words in the document that are not in this list, e.g.,

This is the example. or *This is not an example.*,

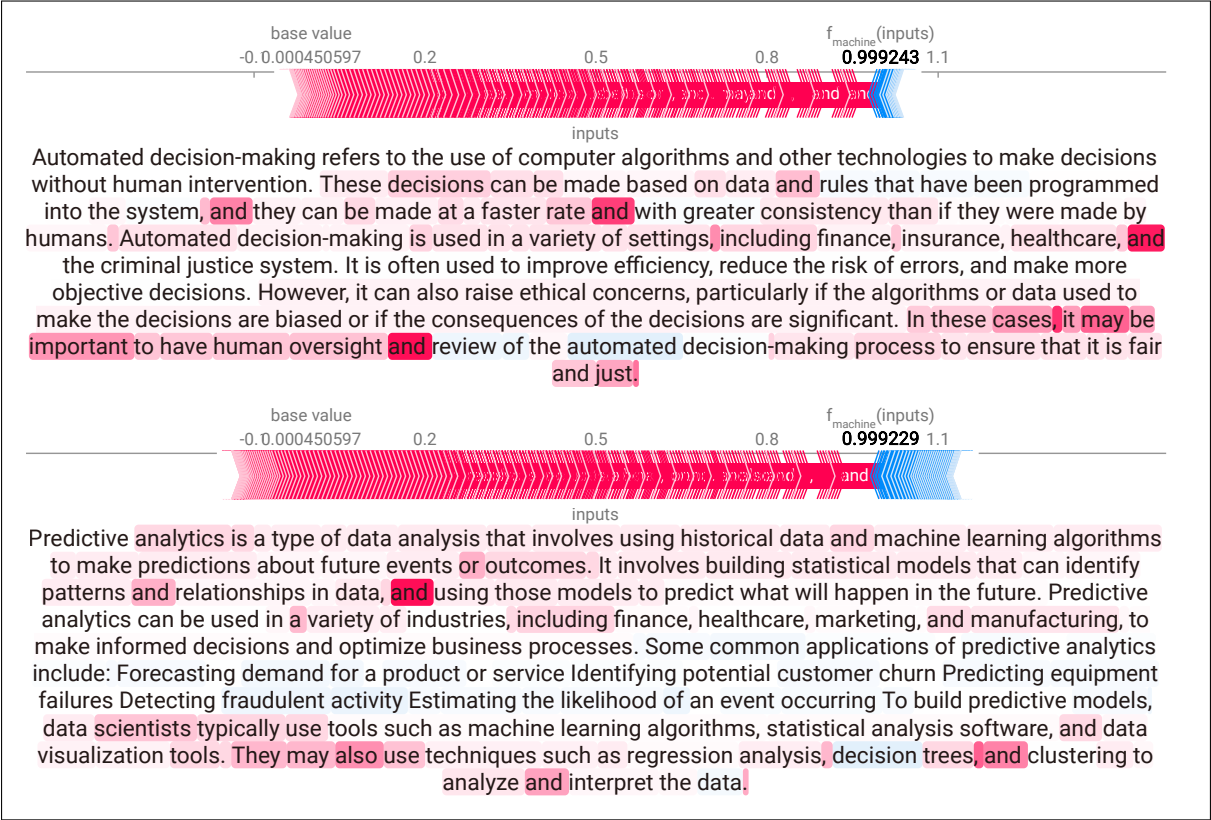
one should expect the detector to still predict *example* 90% of the time.

The visualization is interactive. Clicking on any word in the list updates the rule and documents. Note how the percentage decreases for shorter rules:

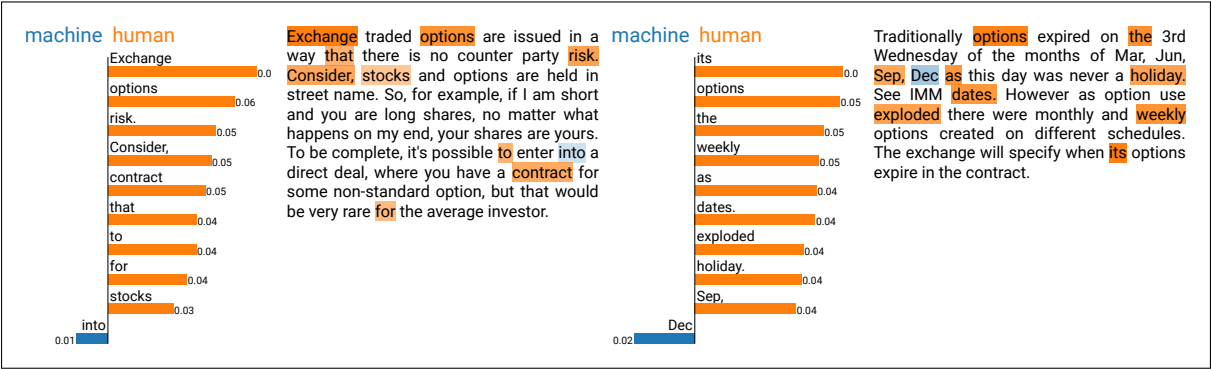
Example	▽ Examples where the Detector predicts example
This is an example.	< 1 2 3 4 5 6 > This was an example.
Explanation of detector prediction	▽ Examples where the Detector DOES NOT predict example
If ALL of these words are in the text: ✓ example This is	< 1 2 3 4 5 6 7 8 > The word example.
The Detector will predict example 60.0% of the time	

Figure 7: Instructions on Anchor shown in Phase 3

(a) Set Guo, selected by SHAP; $y_i = f(d_i) = \text{machine}$



(b) Set DetectGPT, selected by LIME; $y_i = f(d_i) = \text{human}$



(c) Set Solaiman, selected by Anchor; $y_i = f(d_i) = \text{machine}$

Example It is not advisable to ignore tax obligations and start a new business to avoid paying taxes. This could be considered tax evasion, which is a serious offense that can result in criminal charges and significant penalties. Instead, you should try to come to a resolution with the tax authorities to pay the taxes that your company owes. You may be able to negotiate a payment plan or an offer in compromise to resolve the tax debt. Ignoring the problem is likely to make it worse, and could lead to legal action being taken against your company. It is important to seek the advice of a qualified tax professional to help you understand your options and determine the best course of action for your specific situation. They can help you negotiate with the tax authorities and find a solution that works for you and your business. Explanation of detector prediction If ALL of these words are in the text: ☒ and The Detector will predict machine 60.1% of the time	Examples where the Detector predicts machine 1 2 3 4 5 ... 10 You do not trying to ignore tax obligations or start creating new arrangements to avoid paying taxes. This could be considered tax evasion, which is a serious offense that can result in criminal responsibility and civil penalties. Instead, you may try to come to a resolution with the tax authorities to pay the taxes that your company owes. You may be unwilling to negotiate a repayment plan or an offer in compromise to resolve your tax debt. Ignoring a problem is likely to make it worse, and could lead to legal action being taken against your company. It is important to seek professional advice of a qualified tax professional to help you understand your options and determine the best course of action with your specific situation. They can help you negotiate with the tax authorities and find a plan that works for you and your company.	Examples where the Detector DOES NOT predict machine 1 2 3 4 5 ... 10 It is not advisable to ignore tax obligations and start a separate business to avoid paying taxes. This could be considered tax evasion, which is a serious offense that can result in criminal charges and significant penalties. Instead, you should try to come to a resolution with the tax authorities and pay the taxes that your company owes. You may be able to negotiate a alternate plan or an offer of compromise to resolve the tax debt. Ignoring the problem is likely to make it worse, and could lead to legal action being taken against your company . It is important to seek the advice of a qualified tax arbit adviser helping you understand your options and determine the best course of action for your painful situation. They can help you negotiate with the tax authorities and find a solution that works for you and your company.
Example It is important to note that alternative medicine, such as homeopathy and ayurveda, should not be used as a replacement for conventional medical treatment for esophagus disorders. It is important to consult with a medical professional and follow their recommended treatment plan. That being said, some people with esophagus disorders may find relief from symptoms with the use of alternative medicine in addition to conventional treatment. If you are interested in exploring the use of alternative medicine, it is important to discuss this with your doctor and make sure that it is safe and appropriate for your specific condition. It is also important to note that the effectiveness of alternative medicine for esophagus disorders has not been widely studied and there is limited scientific evidence to support its use. Therefore, it is important to be cautious and to carefully consider the potential risks and benefits before trying any alternative treatments. Explanation of detector prediction If ALL of these words are in the text: ☒ and The Detector will predict machine 100.0% of the time	Examples where the Detector predicts machine 1 2 3 4 5 ... 10 It is important to note that alternative treatments, such as homeopathy and ayurveda, should not be used as a replacement for conventional medical treatment for esophagus disorders. It is important to consult with a medical professional to follow a recommended treatment instructions). Having said, some people with esophagus disorders may find relief from symptoms with several use of alternative medicine in addition to conventional medicines. If you are interested in considering the use of alternative medicine, it is important to discuss this with your doctor and make sure that it is safe and suitable for your specific condition . It is also important to note that the efficacy of alternative medicine for esophagus disorders has not been widely studied and there is limited scientific evidence to support this use . Therefore, it is important to be cautious and to carefully consider the potential risks and benefits before trying any alternative treatments.	Examples where the Detector DOES NOT predict machine Could not find any Examples

Figure 8: Pairs of documents shown to users. Users only see explanations for the first document of each pair and annotate the second one.

D Appendix: Complementary Results

	Acc	F1	AUC	TN	FP	FN	TP	ms/evaluation
DetectGPT GPT-2 @ 100 samples	0.502	0.000	0.500	153	0	152	0	15808
DetectGPT pythia-70m @ 100 samples	0.705	0.579	0.704	153	0	90	62	6391
DetectGPT pythia-70m (this paper)	0.744	0.664	0.743	150	3	75	77	898
Solaiman	0.921	0.922	0.921	139	14	10	142	19
Guo	0.990	0.990	0.990	153	0	3	149	18

Table 4: Performance of the detectors. Note that one could obtain better results for DetectGPT when using GPT-2 by adjusting the classification threshold

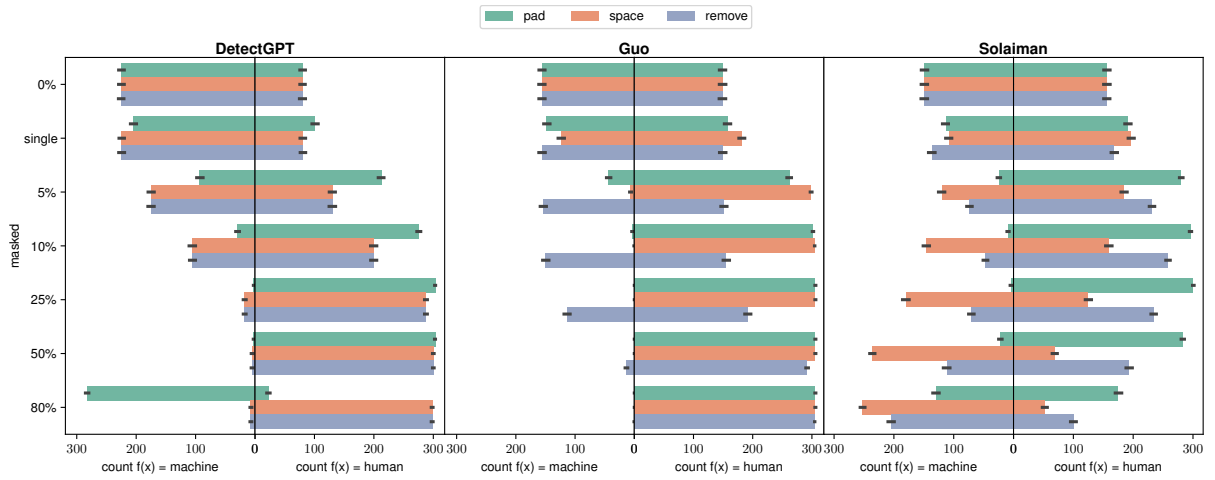


Figure 9: Perturbation Strategy: Shift in detector output at different percentages of tokens randomly removed or replaced. The original dataset is balanced, the counts at 0% of tokens removed serve as a baseline. Mean of 10 runs

(a) Guo			(b) Solaiman			(c) DetectGPT		
Explainer	Score	p	Explainer	Score	p	Explainer	Score	p
LIME	0.605	0.168	LIME	0.402	0.004	Random	0.577	0.000
Random	0.635	0.000	Random	0.484	0.003	Anchor	0.596	0.116
Anchor	0.681	0.000	Anchor	0.492	0.000	LIME	0.631	0.473
SHAP	0.812		SHAP	0.631		SHAP	0.635	

Table 5: Pointing Game: Per detector results. P-values from row-wise binominal tests

(a) Consistency			(b) Continuity		
Explainer	Detector	α	Explainer	Detector	α
SHAP	Solaiman	1.000	SHAP	Guo	0.896
	Guo	1.000		Solaiman	0.812
	DetectGPT	0.084		DetectGPT	0.081
LIME	Solaiman	0.204	LIME	Guo	0.478
	Guo	0.179		Solaiman	0.439
	DetectGPT	0.023		DetectGPT	0.265
Anchor	Solaiman	0.135	Anchor	Guo	0.367
	Guo	0.316		Solaiman	0.135
	DetectGPT	0.097		DetectGPT	0.129
Random	Solaiman	-0.167	Random	Solaiman	-0.137
	Guo	-0.167		Guo	-0.137
	DetectGPT	-0.167		DetectGPT	-0.144

Figure 10: Consistency and Continuity: Per detector results

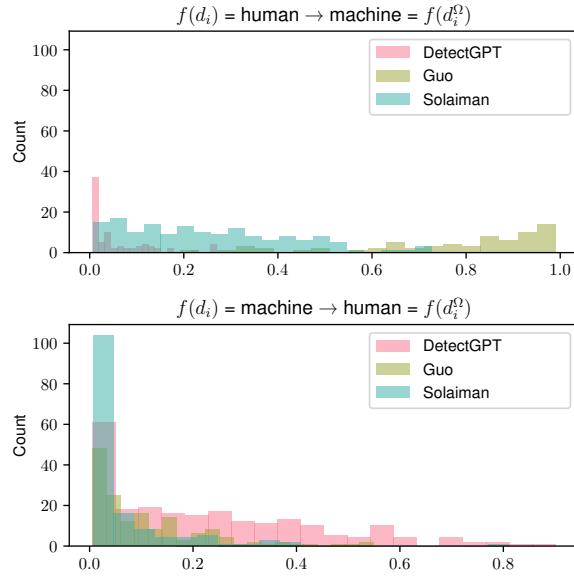
		User Acc without	User Acc with	Change	p
LIME	Solaiman	0.681	0.569	-16.33%	0.057
LIME	DetectGPT	0.792	0.681	-14.04%	0.134
LIME	Guo	0.750	0.681	-9.26%	0.359
Anchor	DetectGPT	0.806	0.750	-6.90%	0.289
SHAP	DetectGPT	0.819	0.806	-1.69%	1.000
SHAP	Solaiman	0.667	0.681	2.08%	1.000
Anchor	Solaiman	0.569	0.583	2.44%	1.000
Anchor	Guo	0.708	0.764	7.84%	0.344
SHAP	Guo	0.778	0.847	8.93%	0.227

DetectGPT	Anchor	Q1	2.15	Solaiman	Anchor	Q1	3.17	Guo	Anchor	Q1	2.39
		Q2	2.04			Q2	3.18			Q2	2.22
		Q3	2.11			Q3	3.15			Q3	2.26
	LIME	Q1	3.62		LIME	Q1	4.00		LIME	Q1	3.17
		Q2	2.97			Q2	3.81			Q2	3.32
		Q3	3.08			Q3	3.65			Q3	3.19
	SHAP	Q1	2.54		SHAP	Q1	3.33		SHAP	Q1	3.29
		Q2	2.56			Q2	3.39			Q2	2.64
		Q3	2.57			Q3	3.29			Q3	2.65

Table 6: Forward Simulation and Perceived Usefulness: Per group results. (3 = neutral, 5 = strongly agree)

		$f(d_i) \rightarrow f(d_i^\Omega)$	n	C_{intra}	C_{inter}
DetectGPT	Random	$h \rightarrow m$	800	0.48	0.49
		$m \rightarrow h$	1990	0.50	0.49
	LIME	$h \rightarrow m$	80	0.39	0.74
		$m \rightarrow h$	199	0.62	0.28
	Anchor	$h \rightarrow m$	80	0.60	0.71
		$m \rightarrow h$	199	0.36	0.56
Guo	SHAP	$h \rightarrow m$	80	0.59	0.59
		$m \rightarrow h$	199	0.59	0.63
	Random	$h \rightarrow m$	90	0.46	0.34
		$m \rightarrow h$	1530	0.49	0.50
	LIME	$h \rightarrow m$	9	0.78	0.44
		$m \rightarrow h$	153	0.52	0.73
Solaiman	Anchor	$h \rightarrow m$	9	1.00	0.22
		$m \rightarrow h$	153	0.66	0.01
	SHAP	$h \rightarrow m$	9	1.00	1.00
		$m \rightarrow h$	153	0.93	0.71
	Random	$h \rightarrow m$	1420	0.48	0.51
		$m \rightarrow h$	1480	0.48	0.51
Solaiman	LIME	$h \rightarrow m$	142	0.79	0.90
		$m \rightarrow h$	148	0.56	0.48
	Anchor	$h \rightarrow m$	142	0.65	0.23
		$m \rightarrow h$	148	0.35	0.05
	SHAP	$h \rightarrow m$	142	0.97	0.94
		$m \rightarrow h$	148	0.88	0.97

(a) Per detector results



(b) Fraction of tokens edited until $f(d_i^\Omega) \neq f(d_i)$

Figure 11: Contrastivity: Per detector results and data on the synthetic datasets